

Electron Orbits from the Signed Electrostatic Force: Covalent, Metallic, and Intermolecular Magnitudes

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Abstract

On the equal-scale locus the electromagnetic force of double spacetime theory is the Coulomb force divided by the interference factor γ_s . The sign of γ_s is the repulsion. An unlike pair attracts where $\gamma_s > 0$ and repels where $\gamma_s < 0$; a like pair does the opposite. No centrifugal term is added to cancel an attraction. A single electron–proton pair then has no force-free radius: the radial force is nonzero wherever γ_s is finite. Two protons and two electrons do, and in the first repulsive shell they do so at exactly one square. Protons sit on one diagonal and electrons on the other, and the electrostatic potential has a positive-definite Hessian there. Like charges attract along the diagonals and unlike charges repel along the sides. The diagonal is $\sqrt{2}$ times the side, the diagonal force is $4.3532 ke^2/\ell^2$, and the stiffness of the diagonal is $8.178 ke^2/\ell^3$. A nearest-neighbor bond that carries only the fraction $2/z$ of that pair has further points where every electrostatic component vanishes. For eight and for twelve neighbors those points are saddles, and the partial forces stand in the ratios 0.04323 and 0.1704 to the covalent diagonal. Two neutral pairs a few side lengths apart keep a force after the monopole cancels. Identifying the side with the Bohr radius makes the diagonal $\sqrt{2} a_0$, against the Born–Oppenheimer separation $1.4011 a_0$ of H_2 . The same layered sign, read in the gravitational channel, is the exclusion of a shared seat: two electron masses placed inside the first gravitational node meet a simple pole. The smallness of Gm_e^2 thins that neighborhood and does not remove the pole.

1 Introduction

The electron-shell section of double spacetime theory writes an electrostatic force whose denominator is the scalar interference factor γ_s [1]. At large separation the factor tends to one and the Coulomb law is recovered. Closer in, γ_s passes through zero on a discrete set of radii and is negative between those nodes. A node is a divergence of the force. The radial potential on the outer side of a node is unbounded below.

A purely attractive Coulomb force does not vanish at any finite radius. The classical circular orbit cancels it by a centrifugal term $L^2/(mr^3)$, the kinematics of a conserved angular momentum. That term is not added here. The outward contribution is already in the signed force. Where γ_s is negative an unlike pair repels, and where γ_s is positive a like pair repels. An electron orbit, in this note, is a locus on which these electrostatic contributions cancel. It is not a balance between Coulomb attraction and a centrifugal barrier.

The note computes that locus for a two-center covalent arrangement, for a nearest-neighbor bond that carries only $2/z$ of an electron pair, and for the residual force between two neutral pairs. A further stability belongs to the gravitational channel. The main text reads the Pauli principle as the first repulsive layer of that channel, generated when two fermions overlap, not as a separate statistical postulate. Applied to electrons, the same layer expels two masses from one orbital seat. The electron mass is small, and the force that expels them is a pole. The equal-scale identification $\alpha = \beta = \ell/r$ is the reading already used for the nuclear and Coulomb layers. The length ℓ is not derived. Bohr’s n^2 rule and the fine-structure constant are not derived. Where a laboratory number enters, it is marked as an input.

2 The signed force

On one axis the interference factor of the dual rotor is

$$\gamma_s = \cosh \frac{\alpha}{2} \cos \frac{\beta}{2} - \sinh \frac{\alpha}{2} \sin \frac{\beta}{2}.$$

The equal-scale locus sets $\alpha = \beta = \ell/r$. In the dimensionless variable $x = \ell/(2r)$,

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x, \quad \gamma'_s(x) = -2 \cosh x \sin x. \quad (1)$$

Thus $\gamma_s(0) = 1$, and on each interval $[n\pi, (n+1)\pi]$ there is one simple zero. The first zero x_1 is the unique root of

$$\tan x = \coth x \quad (2)$$

in $(\pi/4, \pi/2)$. Numerically $x_1 = 0.93755$, so the outermost node lies at

$$\frac{r_1}{\ell} = \frac{1}{2x_1} = 0.53330.$$

The second zero lies at $x_2 = 3.9274$, and the inner edge of the first repulsive shell is $r_2/\ell = 0.12731$. Ordered by decreasing radius, the outer well $r > r_1$ has $\gamma_s > 0$, the shell $r_2 < r < r_1$ has $\gamma_s < 0$, and the sign continues to alternate. The extremum inside the first shell is $\gamma_s(\pi) = -\cosh \pi$, at $r = \ell/(2\pi)$.

The force that increases the separation of two charges is

$$F_r = \frac{kq_1q_2}{\gamma_s(r)r^2}, \quad k = \frac{1}{4\pi\epsilon_0}. \quad (3)$$

Like charges repel where $\gamma_s > 0$ and attract where $\gamma_s < 0$. Unlike charges do the opposite. Both directions are the same formula. Nothing further is added to produce a repulsion.

The Taylor expansion at the origin follows from $\gamma''_s(0) = -2$ and $\gamma'''_s(0) = -4$,

$$\gamma_s(x) = 1 - x^2 - \frac{1}{6}x^4 + O(x^6). \quad (4)$$

Hence

$$\frac{1}{\gamma_s(r)r^2} = r^{-2} + \frac{\ell^2}{4}r^{-4} + \frac{7\ell^4}{96}r^{-6} + O(r^{-8}).$$

The correction to the Coulomb force begins at order $(\ell/r)^2$.

For $r > r_1$ the pair potential that vanishes at infinity is the integral of (3). The substitution $s = \ell/(2x)$ gives $ds/s^2 = -(2/\ell)dx$, and therefore

$$\int_r^\infty \frac{ds}{\gamma_s(s)s^2} = \frac{2}{\ell} \int_0^{\ell/(2r)} \frac{dx}{\gamma_s(x)}, \quad (5)$$

provided the path stays in the outer well. The integral diverges as the upper limit approaches x_1 . Inside a shell the same formula does not continue: the path from infinity would cross a pole. A local force balance inside the shell is still well defined, because (3) requires only $\gamma_s \neq 0$.

3 No centrifugal term

Consider one proton and one electron. The radial force is

$$F_r = -\frac{ke^2}{\gamma_s(r)r^2}.$$

Wherever γ_s is finite and nonzero, F_r is nonzero. In the outer well it is attractive at every finite radius. In the first repulsive shell it is repulsive at every finite radius. There is no radius at which the electrostatic force vanishes, and none is introduced by hand.

The centripetal rearrangement $v^2 = ke^2/(m\gamma_sr)$ of the main paper records what a conserved angular momentum would contribute in the outer well. It is a kinematic identity for a circular Newtonian orbit. It is not an extra force in (3), and it is not used below. The repulsive shell already reverses the electron–proton force. At its quietest point, $x = \pi$,

$$|F| = \frac{4\pi^2}{\cosh \pi} \frac{ke^2}{\ell^2} = 3.4057 \frac{ke^2}{\ell^2}, \quad (6)$$

directed outward. That is the built-in repulsion, evaluated where $|\gamma_s|$ in the first shell is largest and the force is therefore smallest. A one-center electron is not parked on a force-free circle. A force-free locus appears only when several charges are present and their signed contributions cancel.

4 Exclusion as gravitational repulsion

The gravitational channel uses the same interference factor, sourced by the kinematic pair tied to inertial mass rather than by the pair tied to charge [1]. For two electron masses the force that increases their separation is

$$F_G = -\frac{Gm_e^2}{\gamma_{s,G}(r)r^2}, \quad (7)$$

with $\gamma_{s,G}$ the equal-scale factor of that channel and with its own length ℓ_G . Where $\gamma_{s,G} > 0$ the masses attract. Where $\gamma_{s,G} < 0$ they repel. The first zero of $\gamma_{s,G}$ is simple, by the same derivative (1). Approached from inside, F_G therefore has a simple pole and points outward.

The prefactor is small. The CODATA values give

$$\frac{Gm_e^2}{ke^2} = 2.401 \times 10^{-43}. \quad (8)$$

A simple pole is not removed by a small prefactor. Write the inward fractional displacement from the first gravitational node as $r = r_1(1 - \varepsilon)$ with $\varepsilon \downarrow 0$. Then $x - x_1 = x_1\varepsilon + O(\varepsilon^2)$ and

$$|\gamma_{s,G}| = 2x_1 \cosh x_1 \sin x_1 \varepsilon + O(\varepsilon^2) = 2.2260 \varepsilon + O(\varepsilon^2).$$

The outward force exceeds the scale ke^2/r^2 of a nonsingular electromagnetic force at the same radius when $\varepsilon < 1.078 \times 10^{-43}$. The electron mass fixes the width of that neighborhood. It does not decide whether the neighborhood exists.

This comparison assumes that the electromagnetic factor is not itself zero at the same radius. The two channels are distinct phase pairs, so a gravitational node is not required to be an electromagnetic node, and ℓ_G is not fixed by the square of the next section. If the two factors vanished together, their ratio would remain (8) and the electromagnetic piece would dominate. The exclusion read here is the gravitational pole at a node the charge channel does not share.

Placing both electrons on one orbital seat drives their separation through that node. The force (7) is then outward and unbounded, so the shared seat is left at once. The main text proposes that this repulsive layer is the Pauli principle: fermions entering it cannot occupy one seat, without a prior appeal to quantum statistics. That proposal is adopted here for electrons. It is not a derivation of the spin-statistics theorem, and it does not fix ℓ_G .

5 The square and the covalent force

Place protons of charge $+e$ at $z = \pm R/2$ and electrons of charge $-e$ at cylindrical radius ρ , opposite each other in the midplane. The three separations are the proton-proton distance R , the electron-electron distance $r_{ee} = 2\rho$, and the four equal electron-proton distances

$$r_{ep} = \sqrt{(R/2)^2 + \rho^2}.$$

No angular momentum is assigned. Equilibrium means that the axial force on each proton and the cylindrical force on each electron both vanish.

The axial force on the proton at $+R/2$, counted positive when it increases R , and the cylindrical force on one electron, counted positive when it increases ρ , are

$$F_z = ke^2 \left(\frac{1}{\gamma_s(R) R^2} - \frac{R}{\gamma_s(r_{ep}) r_{ep}^3} \right), \quad (9)$$

$$F_\rho = ke^2 \left(\frac{1}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{2\rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (10)$$

The first term of F_ρ is the electron-electron force. In the outer well it is a repulsion and points outward: that outward piece is electrostatic, and it occupies the place a centrifugal term would have occupied. In the repulsive shell $\gamma_s(r_{ee}) < 0$, so the same term points inward, because like charges attract there.

The configuration in which the four charges are the vertices of a square, with the protons on one diagonal and the electrons on the other, is

$$\rho = \frac{R}{2}, \quad r_{ee} = R, \quad r_{ep} = \frac{R}{\sqrt{2}}. \quad (11)$$

On this locus F_z and F_ρ reduce to one expression. With $x = \ell/(2R)$,

$$F_z = F_\rho = \frac{ke^2}{R^2} \left(\frac{1}{\gamma_s(x)} - \frac{2\sqrt{2}}{\gamma_s(\sqrt{2}x)} \right). \quad (12)$$

Both vanish when

$$\gamma_s(\sqrt{2}x) = 2\sqrt{2}\gamma_s(x). \quad (13)$$

The coefficient $2\sqrt{2}$ is greater than one, and on $[0, \pi]$ the factor γ_s decreases. While the diagonal phase x is still less than the first node, $\gamma_s(x)$ is positive and $\sqrt{2}x$ has not yet reached π . The value at the larger phase is then smaller than the value at x , and that value is smaller than $2\sqrt{2}$ times itself. The two sides of the balance do not meet. They still do not meet if only the side has crossed the node, for then one factor is negative and the other is positive. A square whose diagonal lies in the outer well is not an equilibrium.

The balance can hold only after the diagonal has entered the first repulsive shell, so that both the diagonal and the side lie between the two nodes. There both factors are negative: like charges attract along the diagonals, and unlike charges repel along the sides. On that range

$$\gamma_s(\sqrt{2}x) - 2\sqrt{2}\gamma_s(x)$$

increases with x . It is negative at $3/2$ and positive at $8/5$, so it vanishes once, and only once, in the shell. The crossing lies between those two phases. At $8/5$ the side phase is still less than π , and the second node lies beyond π , so the crossing is inside the shell. Its phase is

$$x = 1.55463, \quad \frac{R}{\ell} = 0.32162, \quad \frac{r_{ep}}{\ell} = 0.22742.$$

The two interference factors are $\gamma_s(x) = -2.2208$ on the diagonals and $\gamma_s(\sqrt{2}x) = -6.2812$ on the sides. The repulsion that holds the electrons off the protons is the side force. It is not a centrifugal force.

The partial forces at this root are

$$|F_{\text{diag}}| = \frac{ke^2}{|\gamma_s(x)| R^2} = 4.3532 \frac{ke^2}{\ell^2}, \quad (14)$$

$$|F_{\text{side}}| = \frac{ke^2}{|\gamma_s(\sqrt{2}x)| r_{ep}^2} = 3.0782 \frac{ke^2}{\ell^2}. \quad (15)$$

The diagonal force is the covalent force scale: it is the attractive force along the proton–proton separation, cancelled by the axial part of the side repulsions.

The electrostatic potential, regarded as a function of R and ρ near this root, has Hessian eigenvalues $37.33 ke^2/\ell^3$ and $7.933 ke^2/\ell^3$, both positive. The root is a local minimum. Allowing ρ to follow R , the stiffness of the diagonal is

$$\kappa = 8.178 \frac{ke^2}{\ell^3}. \quad (16)$$

Two further roots of $F_z = F_\rho = 0$ lie in the same shell. Each has Hessian eigenvalues of opposite sign, so each is a saddle. The square is the stable covalent orbit. Its two electrons occupy opposite vertices, a separation R , not one seat. A displacement that puts both on one vertex is the overlap of Section 4: once that separation reaches the first gravitational node, F_G expels them. The electromagnetic Hessian stabilizes the square against small displacements. The gravitational pole forbids the shared seat even though m_e is small. Because the root lies inside the shell, the outer-well integral (5) does not assign it a dissociation energy relative to separated atoms at infinity.

6 Coordination and the metallic force

A nearest-neighbor bond in a monovalent coordination shell of z neighbors is given the electron content $\nu = 2/z$. The dimer is $z = 1$, $\nu = 2$, the square of the previous section. The charge $-\nu e$ is split into two lumps of charge $-(\nu/2)e$ at opposite points of the midplane. The forces (9) and (10) become

$$F_z = ke^2 \left(\frac{1}{\gamma_s(R) R^2} - \frac{\nu R}{2 \gamma_s(r_{ep}) r_{ep}^3} \right), \quad (17)$$

$$F_\rho = ke^2 \left(\frac{(\nu/2)^2}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{\nu \rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (18)$$

Equilibrium is still $F_z = F_\rho = 0$, with no centrifugal term. The outward piece in the outer well is the first term of F_ρ : the lumps repel where $\gamma_s(r_{ee}) > 0$.

For $z = 8$ and $z = 12$ that outer-well system has one root each. All three separations lie outside r_1 , so the signs are the ordinary ones: like charges repel, unlike charges attract. The partial proton–proton force and its ratio to the covalent diagonal (14) are

$$z = 8 : \quad \frac{R}{\ell} = 2.3591, \quad F_{\parallel} = 0.18820 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.04323, \quad (19)$$

$$z = 12 : \quad \frac{R}{\ell} = 1.2666, \quad F_{\parallel} = 0.74188 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.1704. \quad (20)$$

At both roots the Hessian of the potential has eigenvalues of opposite sign. Each is a saddle, not a local minimum. The same is true of the further roots that sit inside the repulsive shell. Reducing the electron content from a full pair to $2/z$ leaves a point where the signed forces cancel, and it does not leave a stable orbit. A crystal in which each ion is surrounded symmetrically can still have a vanishing net force by symmetry. That lattice sum is not the calculation above. An empirical nearest-neighbor distance would fix $F_{\parallel}$ by the proton–proton term alone; it is an input, and it is not required for the ratios.

7 Neutral pairs and the intermolecular force

The stable covalent orbit fixes an electron–proton distance $a = R/\sqrt{2} = 0.22742\ell$. A neutral pair is a proton with an electron displaced by that distance. It is not a force-free one-center orbit: Section 3 has none. It is the side of the square, used as the size of a neutral pair. Two such pairs whose centers are a distance D apart have four intermolecular forces, each of the form (3). If each electron were coincident with its proton, every pair would share one distance and the sum would vanish with the net charge. The monopole cancels. What remains is the displacement a , read with the signed factor γ_s .

Two collinear placements are compared. In the parallel placement both electrons are displaced by $+a$ along the line of centers. In the inward placement the electrons lie between the protons. Attraction is counted positive. F is the net force on the left-hand pair and F_C is the same sum at $\gamma_s = 1$, so that the difference is the torsional excess. In units of the covalent diagonal,

D/a	$F/ F_{\text{diag}} $	$F_C/ F_{\text{diag}} $	$F/ F_{\text{diag}} $	$F_C/ F_{\text{diag}} $
	parallel		inward	
4	+0.5937	+0.1160	+4.537	−0.4010
6	+0.03668	+0.02155	−0.1056	−0.04565
8	+0.008803	+0.006679	−0.01666	−0.01149

At four side lengths the inward electron–electron separation is $2a = 0.4548\ell$, which lies inside r_1 . That pair has $\gamma_s < 0$, so the two electrons attract, and the net inward force is several times the covalent diagonal while the pure Coulomb sum is repulsive. The amplification is the sign change in (3), not an extra interaction. At six and eight side lengths every intermolecular separation is outside the node. Both columns fall. At eight side lengths the parallel residue is 0.88% of the covalent diagonal, and about a quarter of that residue is the torsional excess over F_C .

8 A numerical illustration

The ratio of diagonal to side is $\sqrt{2}$ by (11), with no choice of ℓ . One identification turns the side into the Bohr radius,

$$a = r_{ep} = a_0.$$

The diagonal is then

$$R = \sqrt{2}a_0 = 1.4142a_0,$$

against the Born–Oppenheimer equilibrium separation $1.4011a_0$ of H_2 [3]. The predicted separation is long by a factor 1.009. The identification $a = a_0$ is an input; the factor $\sqrt{2}$ is the square. With the CODATA values of a_0 and of the Hartree energy $E_h = ke^2/a_0$ [2], the same identification gives $\ell = 4.3971a_0$ and

$$|F_{\text{diag}}| = 18.55 \text{ nN}, \quad |F_{\text{side}}| = 13.12 \text{ nN}, \quad \kappa = 150 \text{ N m}^{-1}.$$

The diagonal force is the attractive proton–proton force at the equilibrium. The side force is the repulsive electron–proton force. The stiffness is (16). No dissociation energy is quoted: the equilibrium lies in the repulsive shell, and (5) does not cross the node. The measured dissociation energy of ortho- H_2 from $N = 1$ is 4.463 eV [4]; it is an external number, and it is not a value of this potential.

9 Conclusions

The sign of γ_s is the repulsion in (3). A centrifugal term is not required, and it is not introduced. One electron and one proton have no force-free radius. Two of each have exactly one square in the first repulsive shell, with phase between $3/2$ and $8/5$, diagonal force $4.3532 ke^2/\ell^2$ and stiffness $8.178 ke^2/\ell^3$. The bonds that carry $2/8$ and $2/12$ of an electron pair have electrostatic roots and indefinite Hessians; their partial forces are 0.04323 and 0.1704 of the covalent diagonal. Neutral pairs separated by eight side lengths retain a parallel force 0.0088 of that diagonal, and at four side lengths an inward pair enters the repulsive shell and the force changes by the sign of γ_s .

The square is a local minimum, and its electrons sit on opposite vertices. The gravitational channel supplies the complementary stability: two electron masses driven inside the first gravitational node meet a simple pole, so a shared seat is expelled. The factor $Gm_e^2/(ke^2) = 2.401 \times 10^{-43}$ thins that neighborhood and does not remove it. The node still prevents the outer-well integral from assigning a dissociation energy, and the Bohr spectrum is not implied. One identification, the side equal to a_0 , converts the diagonal into $\sqrt{2}a_0$.

References

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